Aluno: Silas Hoffmann Reis Viana

Tipo	**Tempo base**	**Use para o teste (i = i op 3)**			**Use para o teste (i = i op j)**		
		Soma	*Or*	*Mult*	*Soma*	*Or*	*Mult*
byte	**2715302**	**2778448**	**2778448**	**2904737**	**2904737**	**2904737**	**3094177**
int	**2967884**	**3094180**	**3031033**	**3283619**	**3346761**	**3346761**	**3851935**
float	**3473057**	**12690186**	**XXXXX**	**10608758**	**12942770**	**XXXXX**	**10861342**

	MIPS (ATM328P)					
	Constante (Ex.: i=i op 3)			**Variável (i = i op j)**		
Tipo	*Soma*	*Or*	*Mult*	*Soma*	*Or*	*Multi*
byte	**15.836**	**15.836**	**5.278**	**5.278**	**5.278**	**2,639**
int	**7,918**	**15,836**	**3,167**	**2,639**	**2,639**	**1,131**
	MFLOPS (ATM328P)					
	Constante (Ex.: i=i op 3)			**Variável (i = i op j)**		
Tipo	*Soma*	*Or*	*Mult*	*Soma*	*Or*	*Multi*
float	**0,108**	**XXXXXX**	**0,140**	**0,106**	**XXXXXX**	**0,135**

	CPI					
Tipo	*Soma*	*Or*	*Mult*	*Soma*	*Or*	*Mult*
byte	**1.010**	**1.010**	**3.030**	**3.030**	**3.030**	**6.062**
Int	**2,021**	**1,010**	**5,052**	**6,062**	**6,062**	**14,145**
float	**147,474**	**XXXXXX**	**114,171**	**151,515**	**XXXXXX**	**118,213**

```
long c;
long inicio, fim, tempo;
byte i, j;

void setup() {
  Serial.begin(9600);
}

void loop() {
  i = 1;
  j = 3;

  inicio = micros();

  for(c = 0; c < 1000000; c = c + 1) {
    // --- BASELINE ---
    i = j;

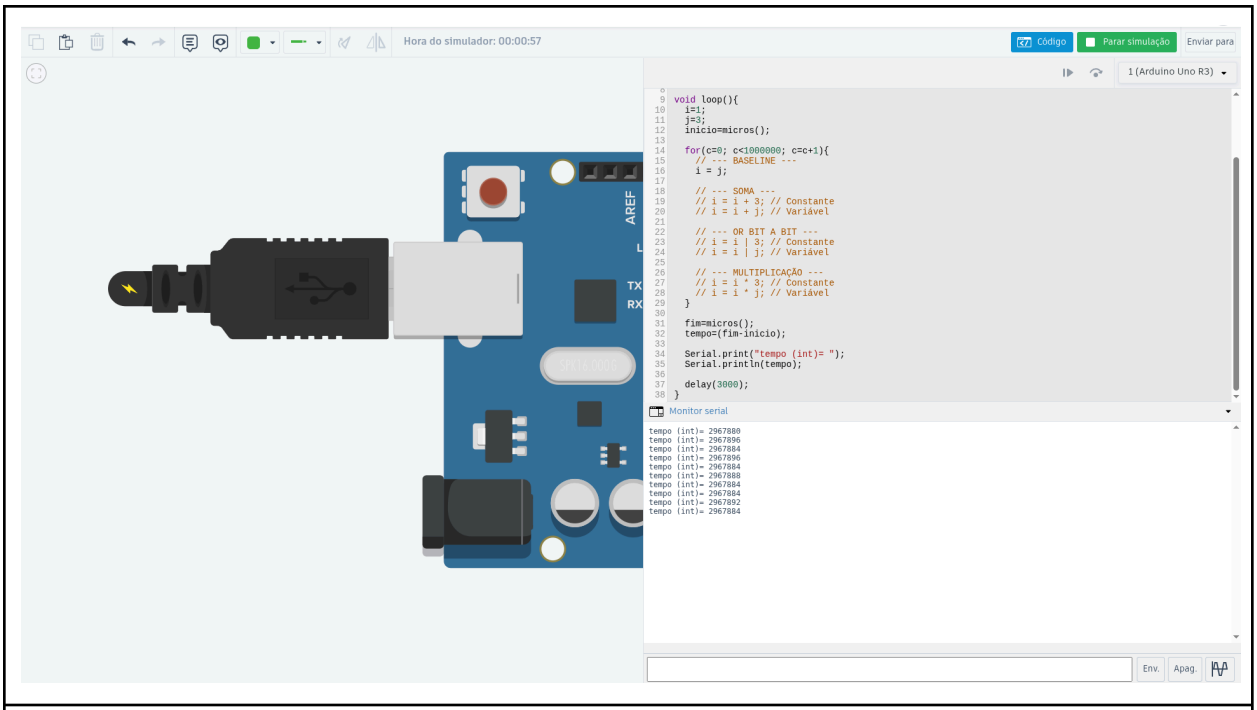
    // --- SOMA ---
    // i = i + 3; // Constante
    // i = i + j; // Variável

    // --- OR BIT A BIT (Pular quando o tipo de dado for float) ---
    // i = i | 3; // Constante
    // i = i | j; // Variável

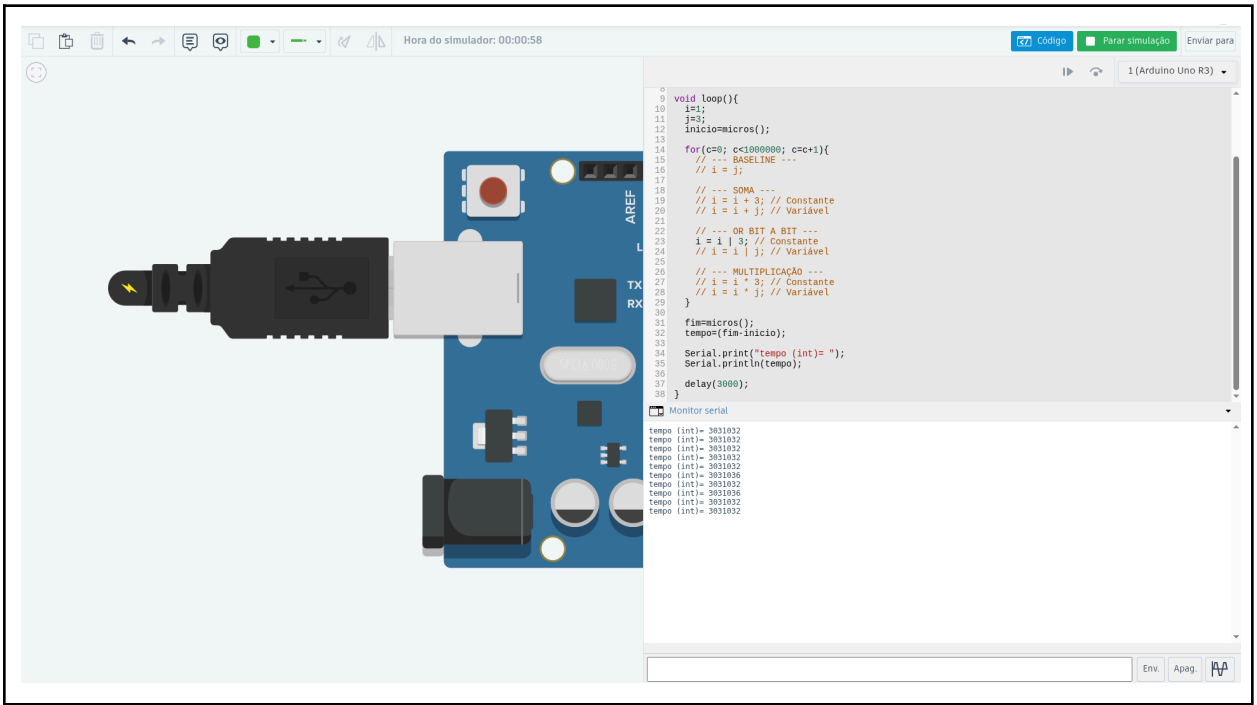
    // --- MULTIPLICAÇÃO ---
    // i = i * 3; // Constante
    // i = i * j; // Variável
  }

  fim = micros();
  tempo = (fim - inicio);
  Serial.print("Tempo medido (us) = ");
  Serial.println(tempo);
  delay(1000);
}
```

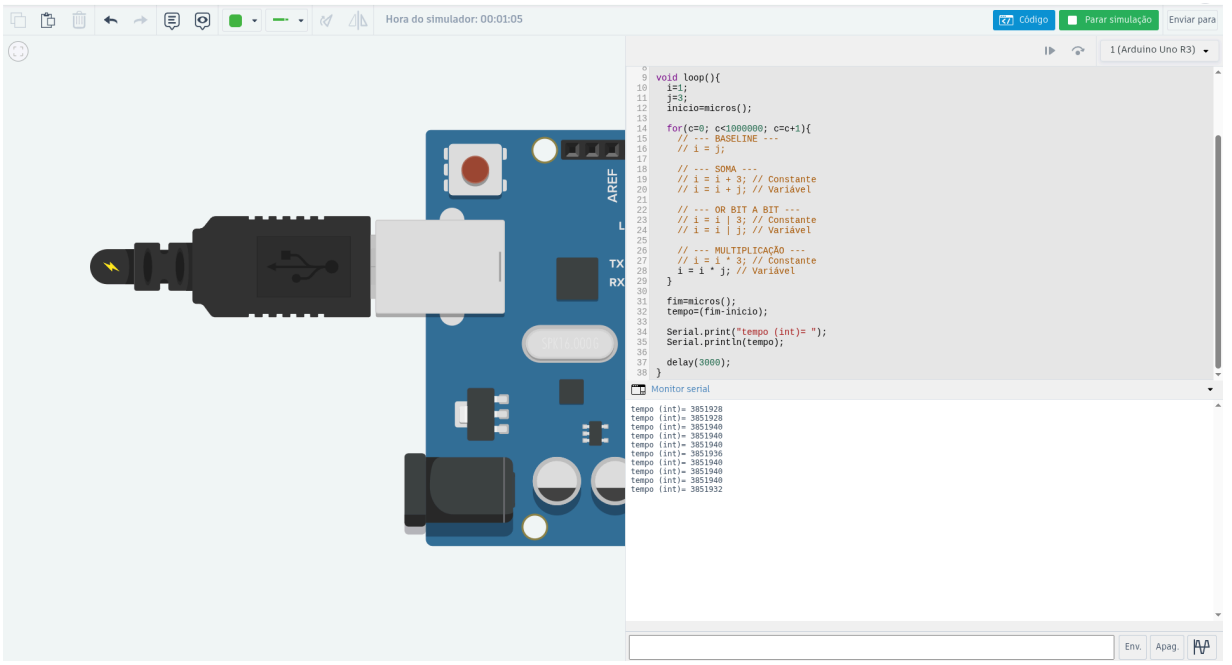
Exemplo de código utilizado (byte)



Teste 10 × (*Int Baseline*)



Teste 10 × (*Int or constante*)



Teste 10 × (*Int mult variavel*)